QNSMARTI:H:QPCODE

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Discipline Specific Elective Course (DSE): MG4DSEBOTA01

AGRIFOOD SCIENCE AND NANOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 h **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any 10 questions.

Each question carries 2 marks

		Taxonomy	CO
1	Name any two major components of agrifood system.	Remember	1
2	Give one example of a food product derived from each of the following sources: a Forestry b. Aquaculture	Remember	1
3	Mention any two areas that highlight the scope of nanotechnology.	Remember	1
4	What are quantum dots? Mention any one application of quantum dots.	Remember	1
5	Give two ways how nanotechnology can improve the quality of food.	Understand	2
6	Explain the role of nanofilters in ensuring the safety of fruit juices	Understand	3
7	Classify the nanomaterials used in food packaging into organic and inorganic types with one example each.	Understand	2
8	State the need for regulations in the application of nanotechnology in food and agriculture.	Understand	3
9	Briefly explain the impact of nano-based products such as Nano Urea and Nano DAP on soil health and crop productivity.	Understand	3
10	Explain how nanotechnology enables precise dosage and targeted application of agricultural inputs.	Understand	4
11	Mention the practical uses of edible nanocoatings and nano-enabled urban agriculture in improving food quality and safety.	Understand	4
12	State the importance of preparing a report to evaluate the impact of nanotechnology applications on agrifood systems.	Understand	3

(2×10=20)

Part B

Short Essay
Answer any 8 questions.
Each question carries 5 marks.

13	Explain the role of nanotechnology in crop production and soil health monitoring.	Remember	1
14	Discuss how nanotechnology enhance food texture and its flavour.	Understand	2
15	Explain the use of portable nanosensor kits made of gold in maintaining food safety.	Understand	3
16	Elaborate how nanobarcodes are used in food quality control for tracking and tracing food products.	Understand	3
17	Discuss why green synthesis is preferred for producing silver nanoparticles and mention its key benefits.	Understand	4
18	Explain the benefits of nanomaterials in food packaging with special reference to oxygen scavenging packaging.	Understand	2
19	Explain the concept of smart packaging using RFID nanotags and mention its advantages in food quality monitoring.	Understand	3
20	Describe the role of government and research organizations such as CeNSE, IFFCO, IARI, and IIT Delhi in the development of nanotechnology for agrifood systems.	Remember	1
21	Outline the role of nanotechnology in modern agrifood systems, including AI-guided automation, nanosensors, edible nanocoatings, and nano-enabled urban agriculture.	Remember	1
22	Write an account of what you learned from a visit to a National or International Centre for Nanoscience regarding nanotechnology applications in agrifood systems.	Remember	1

(5×8=40)

Part 3

Essay
Answer any 1 questions.
Each question carries 10 marks.

23	Describe nano-fertilizers and nano-pesticides, highlighting their advantages over conventional fertilizers and pesticides.	Remember	1
24	Discuss the major challenges of applying nanotechnology in agriculture and allied fields, with reference to toxicity concerns, high cost, scalability issues, lack of awareness, and skill shortage.	Understand	3

(10×1=10)